Retrieval metrics

A retrieval evaluation set contains questions and manually verified relevant source IDs. Recall at K is the number of distinct relevant IDs in the first K results divided by the number of annotated relevant IDs. Several chunks from the same relevant page count once in the numerator when the relevance unit is a page, but they still occupy retrieval ranks.

Reciprocal rank is one divided by the rank of the first relevant result. If no relevant result appears within the evaluated depth, reciprocal rank is zero. Mean reciprocal rank averages this value across questions. DocuSense reports MRR at depth five, called MRR@5, rather than an unrestricted full-ranking MRR.

Macro averaging gives each question equal weight. Incomplete relevance annotations can underestimate recall, while easy development questions can overestimate generalization. Always report the relevance unit, cutoff depth, question count, corpus identity, and whether the questions were used for tuning.

Experiments and reproducibility

An ablation changes a controlled design choice to understand its contribution. Compare TF-IDF and dense retrieval while keeping questions and page labels fixed. Next vary chunk size and overlap, and record indexing time, query latency, Recall@K, and MRR@5. Do not describe development-set improvements as held-out test gains.

Track the model name, dependency versions, random seed, and PDF content hashes. Separate cold model-loading time from warm query latency. A warm embedding cache can make later index builds faster than the first one. The order of experiments therefore matters for timing comparisons.

Reserve independent questions and additional documents for a final evaluation. Avoid choosing all settings by repeatedly consulting a small final test set. State hardware and execution conditions before comparing latency measurements from different machines. Reproducible code does not guarantee identical wall-clock timings.

Exploratory topics

K-Means groups vectors by minimizing squared Euclidean distance to cluster centers. Fix a random seed and use multiple initializations for reproducibility. A representative passage can be selected as the chunk closest to its cluster center. Descriptive keywords summarize frequent terms within the cluster; they are not supervised class labels.

Principal component analysis projects high-dimensional embeddings onto directions of largest variance. A two-dimensional plot helps inspect broad patterns, but projection can distort distances. Nearby points in the plot are not proof of identical topics. The original embedding space remains the basis for clustering.

Silhouette compares within-cluster cohesion with separation from other clusters. It requires at least two clusters and fewer clusters than observations. Compute the score only when valid, and use a bounded sample for larger datasets. High separation does not prove that clusters correspond to useful human categories.